ZEHUNGER SOLUTIONS

Circular Economy ERP

Phase One Walkthrough Guide

From an empty system to a delivered, fully traced sale

A step-by-step guide for presenting the Zehunger Solutions Circular Economy ERP, built from a complete live walkthrough recorded on 30 October 2026.

- Step 0 — Set up the farm and the team
- Steps 1–5 — Waste, substrate, batch, rearing, harvest
- Steps 6–7 — Drying and frass curing
- Step 8 — Quality control and release
- Step 9 — Sell and deliver
- Step 10 — The owner's view: trace, dashboard, reports

How to use this guide

This guide follows one real batch of black soldier fly (BSF) larvae through the whole system, from the moment organic waste arrives at the farm to the moment a customer signs for dried larvae and frass fertiliser. Every screen, reference number and figure in it comes from that walkthrough.

- **Presenting live:** follow Steps 0 to 10 in order. Each step says who does it, where to click, what to type, what the system checks, and what to point out to the audience.
- **Presenting from slides:** the PowerPoint version follows the same steps, with speaker notes on every slide.
- **Training staff:** each step can be read on its own; the "What the system checks" boxes explain the rules a user will meet.

Tip: Rehearse on the local demo or development system first. On the live site, Step 0 (set-up) must be done before any data can be entered.

The chain at a glance

Organic waste becomes two products. The system tracks both, all the way to the customer.

Stage	What happens	Reference created
1. Waste in	Organic waste is collected, weighed and checked	Waste lot `WL-`
2. Substrate	Clean waste is prepared into larvae feed	Substrate batch `SUB-`
3. Production	Larvae are reared on the substrate in bio-pods	Production batch `BSF-`
4. Rearing	Feeding, monitoring and health are recorded	Events `EVT-`
5. Harvest	Larvae are harvested; frass is sieved off	Harvest `HRV-`, frass lot `FRS-`
6. Drying	Wet larvae are dried and weighed	Drying batch `DRY-`
7. Curing	Frass is cured into fertiliser	(on the frass lot)
8. Quality	Each product is checked, approved and released	Checks `QCK-`/`FQC-`, approvals `QCA-`, product lots `LOT-`
9. Sales	Orders, stock reservation, dispatch, delivery	Orders `ORD-`, deliveries `DEL-`
10. Insight	Trace, dashboard, Batch 360 and reports	—

Key message for the audience: every kilogram is linked to where it came from. A complaint about one bag of feed can be traced back to the market the waste came from, the batch that ate it and the officer who passed it.

Who does what

The system gives each person only the screens and actions their job needs.

Role	Example person	What they do in the walkthrough
Super admin (owner)	Zihanga Owner	Sets up farms, stores and accounts; sees everything: trace, dashboard, Batch 360, reports, money

Role	Example person	What they do in the walkthrough
Operator (farmer)	Peter	Records waste, substrate, batches, feeding, monitoring, health, harvest, drying and frass curing
QC officer	Daniel Otieno	Checks dried larvae and frass, approves or rejects, releases approved product into stock
Sales officer	Brian Mwangi	Adds customers, takes orders with prices, reserves stock, dispatches, confirms delivery
Viewer	—	Read-only access, for example for investors

Note: operators see only the farms, sites and facilities they are assigned to. Orders, customers and money are visible only to farm-level users.

The example data used in this guide

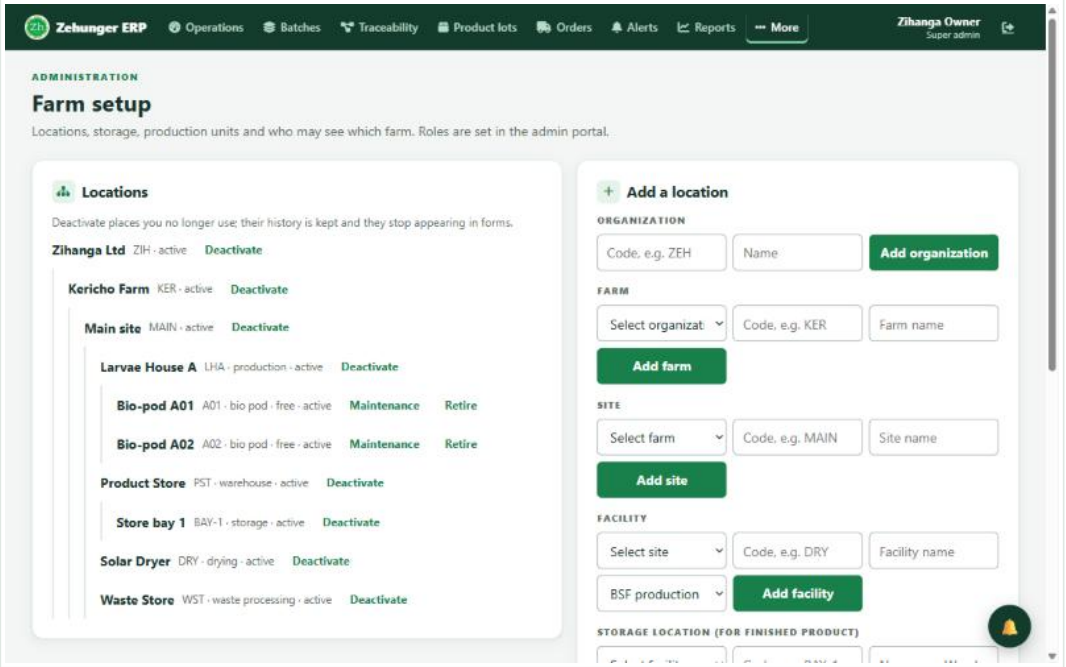
Record	Reference	Key figures
Waste lot	WL-2026-00001	150 kg vegetable waste, Kericho Municipal Market
Waste lot	WL-2026-00002	45 kg, quarantined (not clean enough to use)
Substrate batch	SUB-2026-00001	120 kg prepared from WL-2026-00001
Production batch	BSF-2026-00001	100 kg substrate, bio-pod A01, harvest limit 40 kg
Harvests	HRV-2026-00001, HRV-2026-00002	36 kg + 3 kg wet larvae
Frass lots	FRS-2026-00001, FRS-2026-00002	52 kg + 8 kg separated; 45 kg + 7 kg cured
Drying batches	DRY-2026-00001, DRY-2026-00002	12 kg + 1 kg dried larvae (about 3:1)
Product lots	LOT-2026-00001 to LOT-2026-00004	13 kg dried larvae, 52 kg frass fertiliser
Customer order	ORD-2026-00001	Mama Grace Agrovet: 10 kg larvae + 40 kg frass, KES 3,800
Delivery	DEL-2026-00001	50 kg, received by Grace Chebet

STEP 0

0 Set up the farm and the team

Who: Super admin (owner) · **Where:** *More* → *Farm Setup* (`/setup`) and the admin portal

Before anyone can record work, the farm must exist in the system. Locations are arranged as a tree: **Organization → Farm → Site → Facility → Storage location / bio-pods**.



Farm Setup: the location tree and the forms to add each level

Do this, in order:

1. **Organization:** for example *Zihanga Ltd* (code `ZIH`).
2. **Farm:** *Kericho Farm* (code `KER`).
3. **Site:** *Main site* (code `MAIN`).
4. **Facilities** on that site, one per kind of work:
   - *Waste Store* (waste processing) — where waste arrives
   - *Larvae House A* (BSF production) — where larvae are reared
   - *Solar Dryer* (drying)
   - *Product Store* (warehouse)
5. **Storage location** in the Product Store: *Store bay 1* (`BAY-1`). Released product is received here.
6. **Bio-pods** in Larvae House A: *A01*, *A02*.
7. **Accounts** in the admin portal: one per person, with the right role (operator, QC officer, sales officer), and give each one access to Kericho Farm.

What to point out:

- Places can be **deactivated** but never deleted once used, so history is always kept.
- Self-registration is switched off: **only an administrator creates accounts.**

Live site note: the live database was cleared for the presentation. It keeps the four user accounts but has no farm. Do Step 0 on the live site first (or present on the local demo), and give the operators access to the new farm.

Before you start: signing in and getting around

Who: everyone · **Where:** the landing page (`/`) → *Enter Admin Portal* or *Enter Farmer Portal*

- The **landing page** offers the Admin Portal (office and management) and the Farmer Portal (phone-friendly daily work). The presentation guides are also downloadable from here.
- After signing in, each role lands on its own start page: the owner on the **admin portal**, operators on the **Farmer Portal** (their batches and daily tasks), QC and sales officers on the **Operations dashboard**.
- The **top bar** links the main areas: Operations, Batches, Traceability, Product lots, Orders, Alerts, Reports and More.
- **Data entry** (`/dashboard`) has a sidebar of sections: Waste Management, Larval Production, Drying Process, Frass & Fertiliser, Facility, Hatchery and Records.

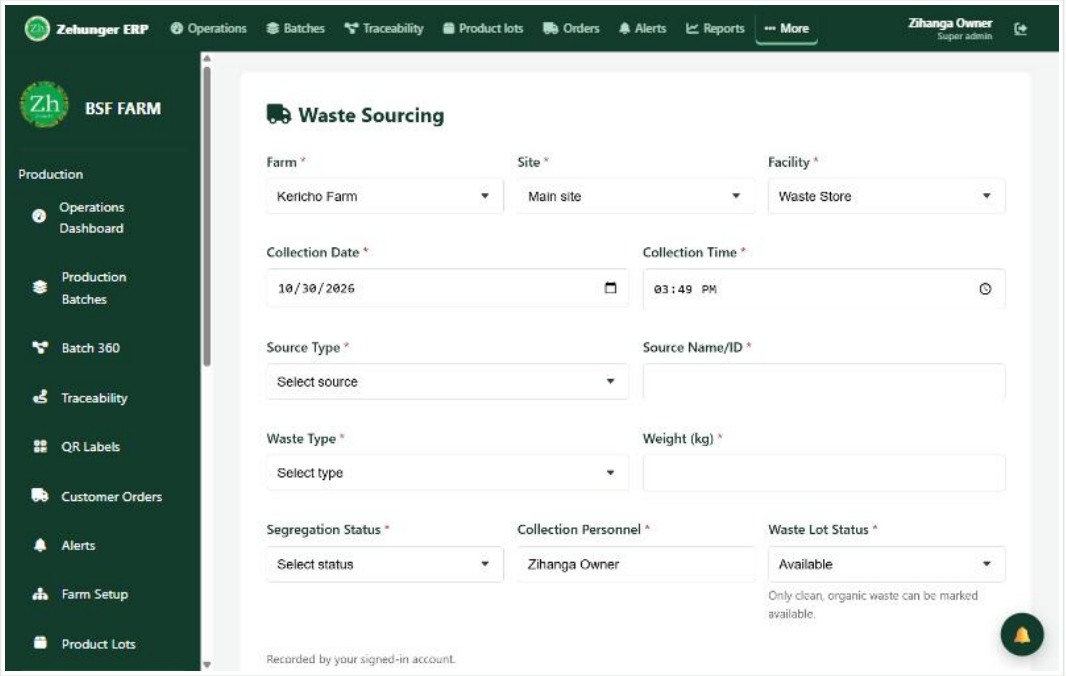
How the data-entry forms behave (worth showing once):

- Each section shows **one form at a time**. Use the arrows or the dots at the bottom to move between forms.
- Dates default to **today** and times to **now**; future dates are refused.
- Name fields default to the **signed-in person**; each form remembers the last farm, site and facility used.
- Problems are shown **under the field as you type**, and the form will not save until they are fixed.
- After saving, a green confirmation appears **at the bottom of the screen and under the button**, with the new reference number.
- Each form lists its **last 5 entries** underneath, so the user can see their work was recorded.

STEP 1

1 Record the waste coming in

Who: Operator · **Where:** *Data entry → Waste Management → Waste Sourcing*



Waste Sourcing form

Fill in:

Field	Example
Farm / Site / Facility	Kericho Farm / Main site / Waste Store
Collection date and time	today, now (filled in)
Source type and name	Market — *Kericho Municipal Market* (also: restaurant, brewery, food industry, household, farm)
Waste type	Vegetable
Weight (kg)	150
Segregation status	Organic
Collection personnel	filled in with the signed-in name
Waste lot status	Available

Result: ✓ *WL-2026-00001* — 150 kg available for substrate.

A second lot to show the quality gate: record 45 kg that arrived mixed with plastic, and set it to **Quarantined**. It becomes *WL-2026-00002* and will **not** appear when preparing substrate.

What the system checks:

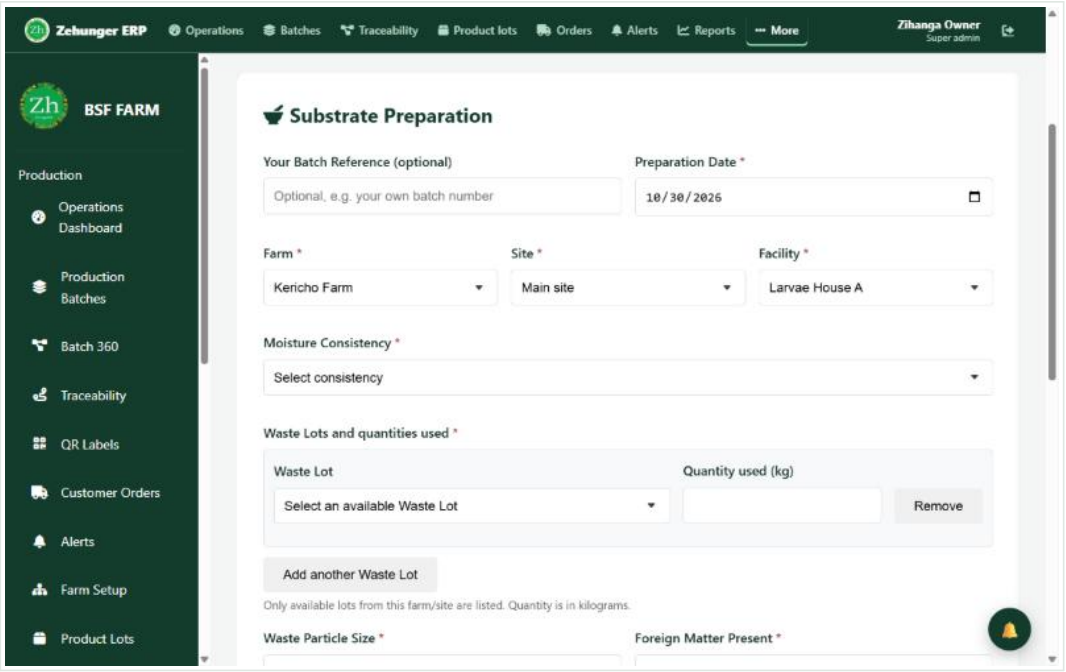
- Only clean, organic waste can be marked *Available*.
- Weights must be positive; dates cannot be in the future.

- The other Waste Management forms (Storage Records, Processing Records, Waste Monitoring) log what happens to waste while it waits.

STEP 2

2 Prepare the substrate

Who: Operator · **Where:** *Data entry → Larval Production → Substrate Preparation*



Substrate Preparation form

Fill in: preparation date, Kericho Farm / Main site / Larvae House A, moisture consistency, then **Waste lots and quantities used**: pick *WL-2026-00001* and enter **120 kg**. Add particle size and whether foreign matter was found.

Result: ✓ *SUB-2026-00001* — 120 kg of substrate. WL-2026-00001 now shows **30 kg left**.

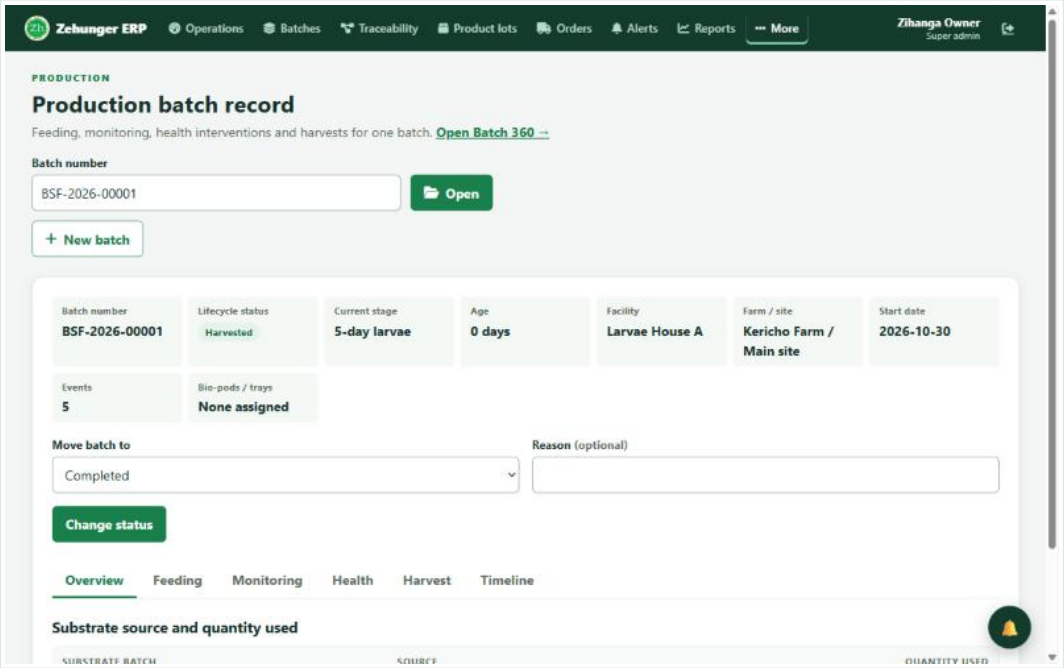
What the system checks:

- Only **available** waste lots on the same farm are offered (the quarantined lot is not).
- You cannot use more kilograms than the lot has left.
- Several waste lots can go into one substrate batch; each link is stored for traceability.

STEP 3

3 Start a production batch

Who: Operator · **Where:** *Batches → Production batch record → New batch* (`/production_batch.html`)



Production batch record

Fill in: facility *Larvae House A*, start date, expected harvest date, bio-pod *A01*, substrate *SUB-2026-00001* with **100 kg** issued, and a **harvest limit of 40 kg**.

Result: ✓ *BSF-2026-00001*, status *Planned*. SUB-2026-00001 now has 20 kg left.

Moving the batch through its life (the *Move batch to* control on the same page):

Planned → Active → Monitoring → Ready for harvest → Harvested → Completed
(Paused and Health alert are available when needed.)

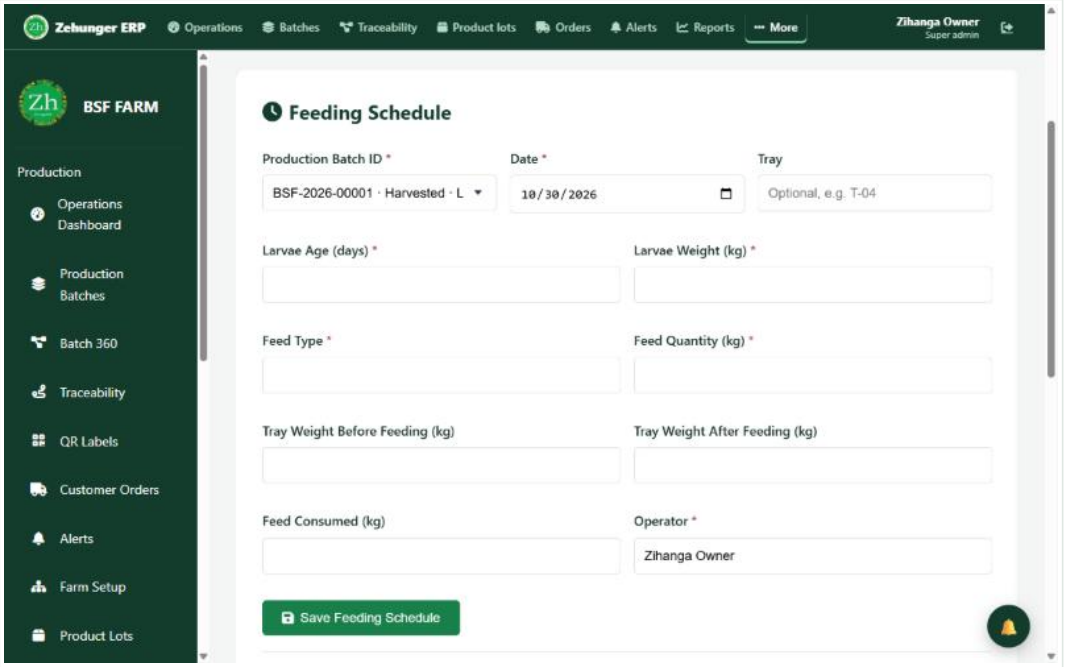
What the system checks:

- The harvest limit caps how much can ever be harvested from the batch.
- A batch must be *Ready for harvest* before a harvest can be recorded.
- Every status change is kept in the batch history with who did it and why.

STEP 4

4 Record daily rearing

Who: Operator · **Where:** *Data entry → Larval Production* (Feeding Schedule, Environmental Monitoring, Health & Intervention)



Feeding Schedule with the production batch picker

Form	What to enter	Example result
Feeding Schedule	batch, larvae age, larvae weight (kg), feed type, feed quantity (kg), consumed (kg), tray	EVT-000001
Environmental Monitoring	temperature (°C), humidity (%), ammonia	EVT-000002
Health & Intervention	issue seen, severity, action taken, follow-up	EVT-000003

What to point out:

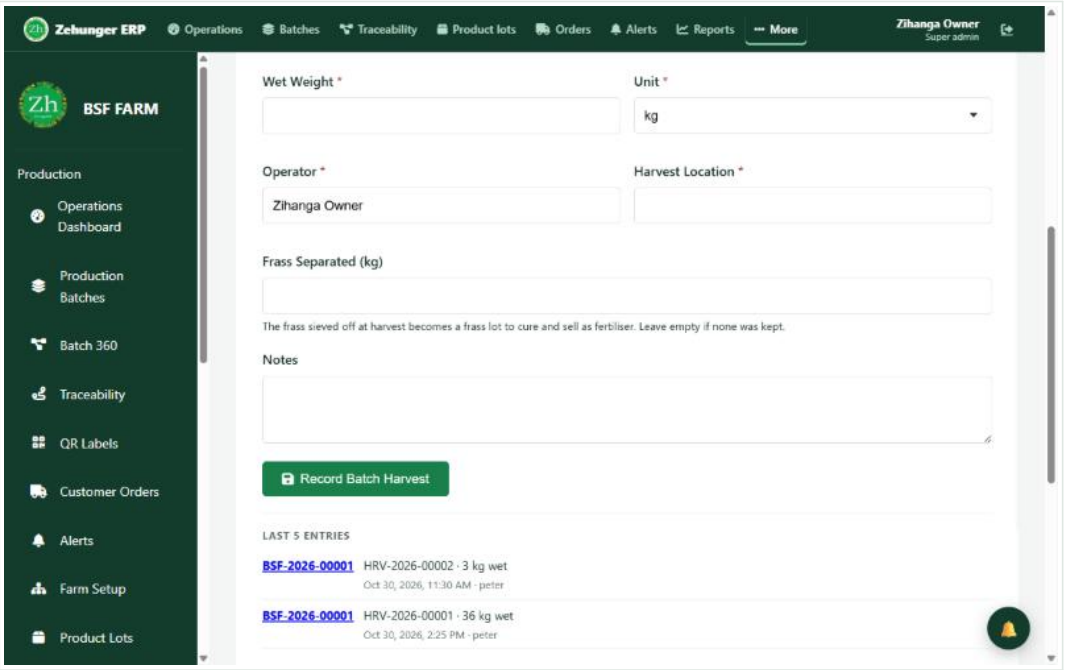
- The **Production Batch** field is a picker of real batches, so events always attach to the right batch.
- Everything is entered **in kilograms**, one unit for every weight.
- Open health issues, harvests that are due or overdue, QC waiting or failed, low or quarantined stock and late orders raise **alerts** (bell icon, top right) for the people responsible.

STEP 5

5 Harvest the larvae and separate the frass

Who: Operator · **Where:** *Data entry → Larval Production → Harvest & Yield* (or the batch record)

First move BSF-2026-00001 to **Ready for harvest**.



Harvest & Yield form with the Frass Separated field and the last entries

Fill in: batch *BSF-2026-00001*, harvest date and time, **wet weight 36 kg**, stage *prepupae*, operator, harvest location, and **Frass Separated: 52 kg**.

Result:

- ✓ *HRV-2026-00001* — 36 kg wet larvae.
- ✓ *FRS-2026-00001* — 52 kg frass, automatically started **curing**.

A second pick from the same batch (3 kg larvae, 8 kg frass) created *HRV-2026-00002* and *FRS-2026-00002*. The batch was then moved to **Harvested**.

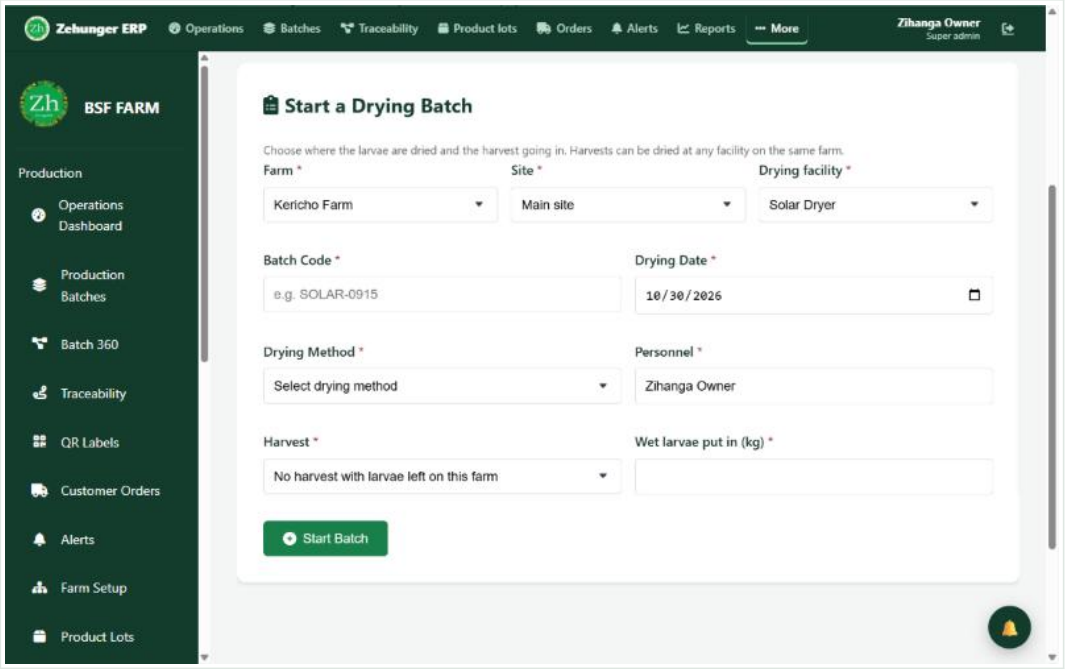
What the system checks:

- Total harvest can never exceed the batch's harvest limit (40 kg here; 39 kg used).
- Frass is the harvest's **second product**: it is not waste, it becomes fertiliser for sale.

STEP 6

6 Dry the larvae

Who: Operator · **Where:** *Data entry → Drying Process*



Start a Drying Batch

The drying forms, in order:

1. **Start a Drying Batch:** farm and site, drying facility *Solar Dryer*, batch code, method (*Sand drying, Sun drying, Solar dryer, Drum dryer, Roasting*), personnel, harvest *HRV-2026-00001* and **36 kg** of wet larvae in.
2. **Add a Harvest to a Drying Batch:** only while drying has not started.
3. **Move a Drying Batch to its Next Stage:** Draft → Received → Drying → Waiting for QC.
4. **Record Dried Output:** weigh what comes off the dryer — **12 kg at 8% moisture**.

Result: ✓ *DRY-2026-00001*: 36 kg wet in, 12 kg dry out (**3:1, 33.3% yield**), now *Waiting for QC*.

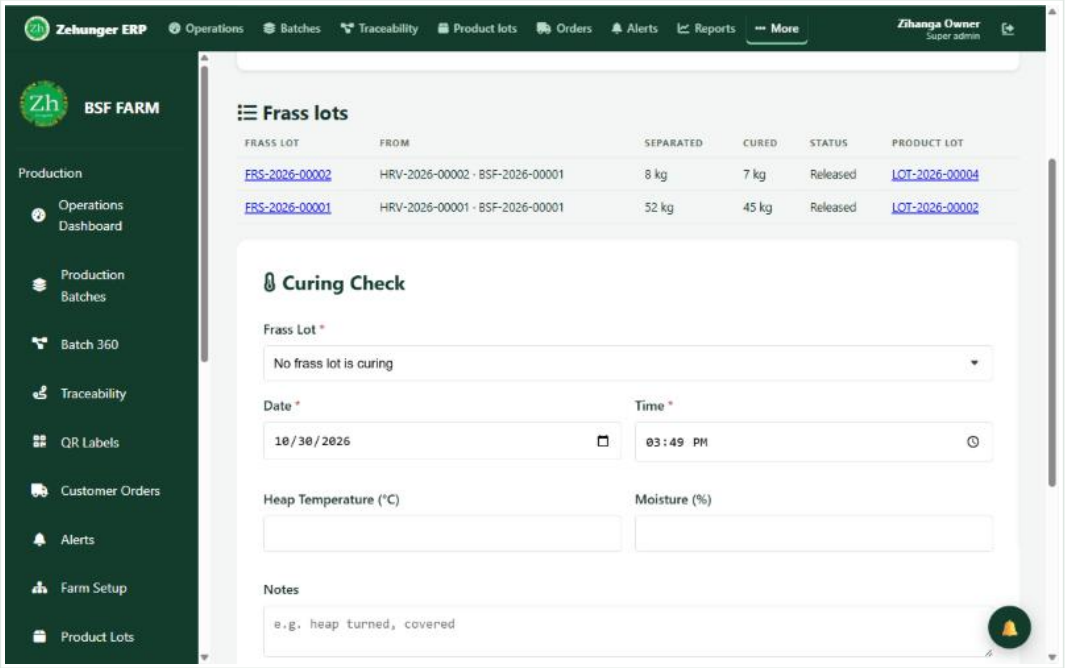
What to point out:

- The guideline box at the top of the section shows the expected ratios (for example 3:1 wet to dry).
- A drying batch can take harvests from several production batches; traceability keeps every link.

STEP 7

7 Cure the frass

Who: Operator · **Where:** *Data entry → Frass & Fertiliser*



Frass lots table and the Curing Check form

The **Frass lots** table at the top shows every lot and where it is: *Curing → Waiting for QC → Approved → Released*.

1. **Curing Check:** while the heap cures, record heap temperature and moisture (for example 45 °C, 32%). At least one check is required.
2. **Finish Curing:** enter the **cured weight** (52 kg became **45 kg**) and the date.

Result: FRS-2026-00001 is cured and **waiting for a quality check**.

What the system checks:

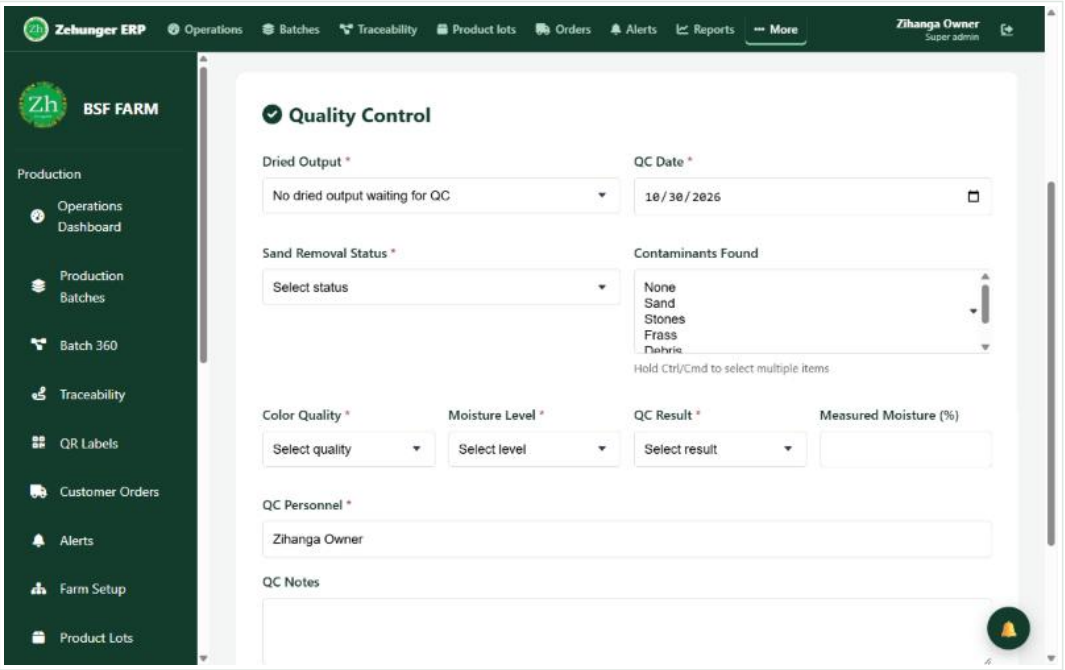
- Curing cannot be finished without at least one curing check.
- The cured weight cannot be more than the frass separated.

STEP 8

8 Quality control and release

Who: QC officer (Daniel) · **Where:** *Drying Process* (forms 6–8) and *Frass & Fertiliser* (forms 3–5)

Each step is its own short form with its own Save button: **check → decide → release.**



Quality Control form

Dried larvae

Form	What Daniel enters	Result
Quality Control	the dried output; sand removal (*Complete*); contaminants (*None*); colour (*Good*); moisture level (*Dry*); measured moisture **8%**; result **Passed**	QCK-2026-00001
QC Decision	the check; decision **Approved**; comments	QCA-2026-00001 — the drying batch completes
Release Product Lot	the approved output; product type *Dried larvae*; store (filled in: *Store bay 1*)	**LOT-2026-00001** — 12 kg in stock

Frass fertiliser

Form	What Daniel enters	Result
Frass Quality Check	the frass lot; maturity **Mature** (dark, crumbly, earthy); moisture **30%**; no contaminants; result **Passed**	FQC-2026-00001
Frass QC Decision	the check; **Approve for sale**; reason	FRS-2026-00001 approved
Release Fertiliser to Stock	the approved lot; store (filled in)	**LOT-2026-00002** — 45 kg frass fertiliser

The second batch went through the same three forms: LOT-2026-00003 (1 kg dried larvae) and LOT-2026-00004 (7 kg frass fertiliser).

What the system checks (strong points for the presentation):

- **Nothing reaches stock without a passed check *and* an approval.** The database itself refuses any product lot without one.
- **Every output of a drying batch is settled** (approved or rejected) before the batch completes, so no output is ever left behind.
- Only the **latest** check on an output can be decided; an old check cannot be approved after a newer one failed.
- **Rejecting needs a reason.**
- *None* cannot be ticked together with a contaminant.
- Frass can pass **only when mature and clean**.
- Product type is a fixed choice, so released stock always matches what sales can order.
- The store list shows **only stores on the product's farm**, and picks it when there is just one.

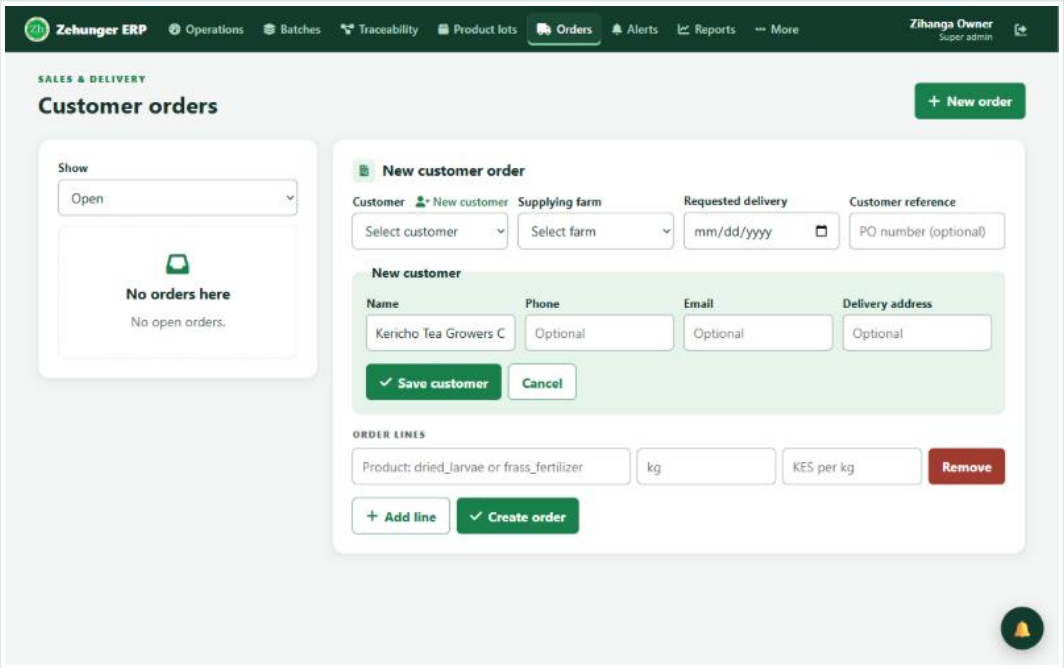
STEP 9

9 Sell and deliver

Who: Sales officer (Brian) · **Where:** *Orders* (`/orders`)

9.1 Add the customer

Click **New order**, then **New customer** beside the Customer label.



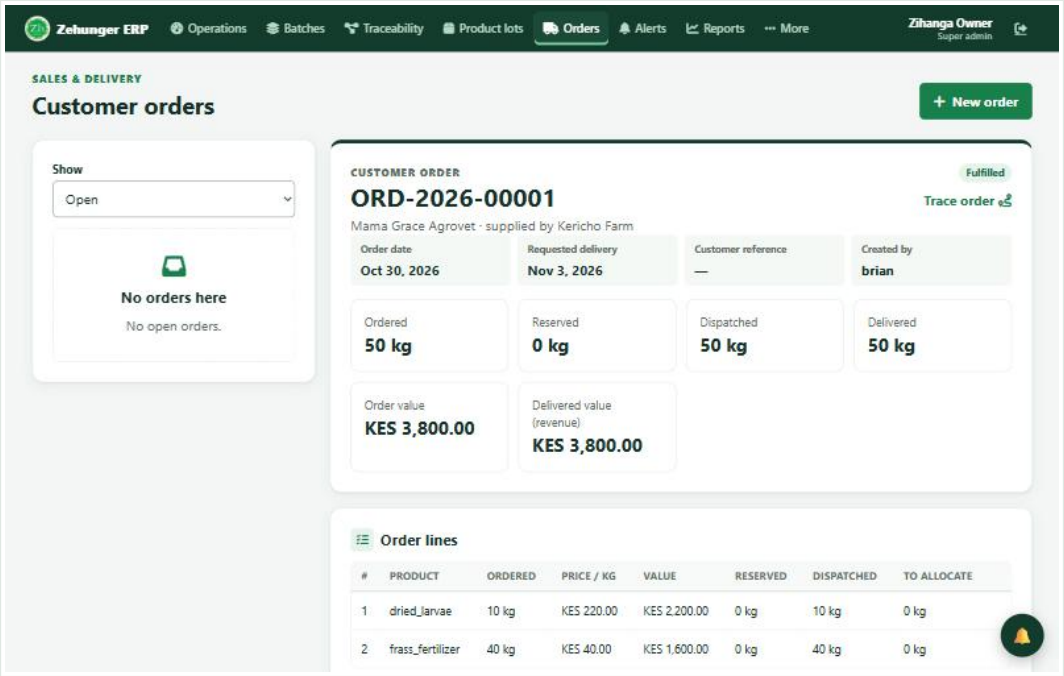
New customer panel inside the order form

Enter the name (*Mama Grace Agrovet*), phone, email and delivery address (*Moi Highway, Kericho town*), then **Save customer**. The customer is created and picked for the order.

9.2 Create the order

Field	Example
Supplying farm	Kericho Farm
Requested delivery	3 Nov 2026
Line 1	`dried_larvae` — 10 kg at KES 220/kg
Line 2	`frass_fertilizer` — 40 kg at KES 40/kg

The product suggestions show **live free stock** (13 kg dried larvae, 52 kg frass). **Create order** → ✓ *ORD-2026-00001*, value **KES 3,800**.



The order with its totals

9.3 Reserve stock

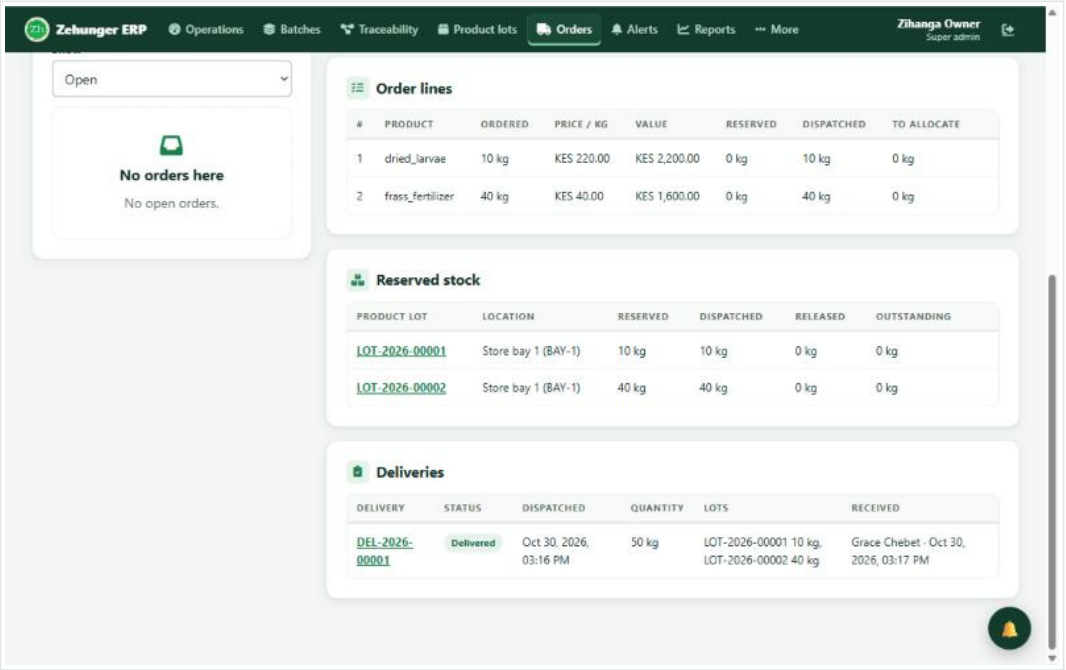
On each line, **Allocate stock** lists free, QC-released lots on the farm, **oldest first**, with suggested amounts. Brian reserves 10 kg from LOT-2026-00001 and 40 kg from LOT-2026-00002. Reserved stock is held: no other order can take it.

9.4 Dispatch

The **Dispatch a delivery** form is filled in with the reserved amounts. Add the vehicle (*KDA 123B pickup*) and driver (*Joseph Kiprono*), then **Dispatch** → ✓ *DEL-2026-00001*. The 50 kg leaves Store bay 1 as recorded stock movements.

9.5 Confirm delivery

When the goods arrive, **Confirm delivered** and enter who received them (*Grace Chebet*). Delivered becomes 50 kg and **revenue becomes KES 3,800**.



Reserved stock and the delivered delivery

What to point out:

- Revenue is counted **only when the customer confirms receipt**: delivered kg × the agreed price.
- *Fulfilled* means everything ordered has been shipped; the delivery itself shows *Delivered*.
- **Cancel order** returns any reserved stock to inventory.

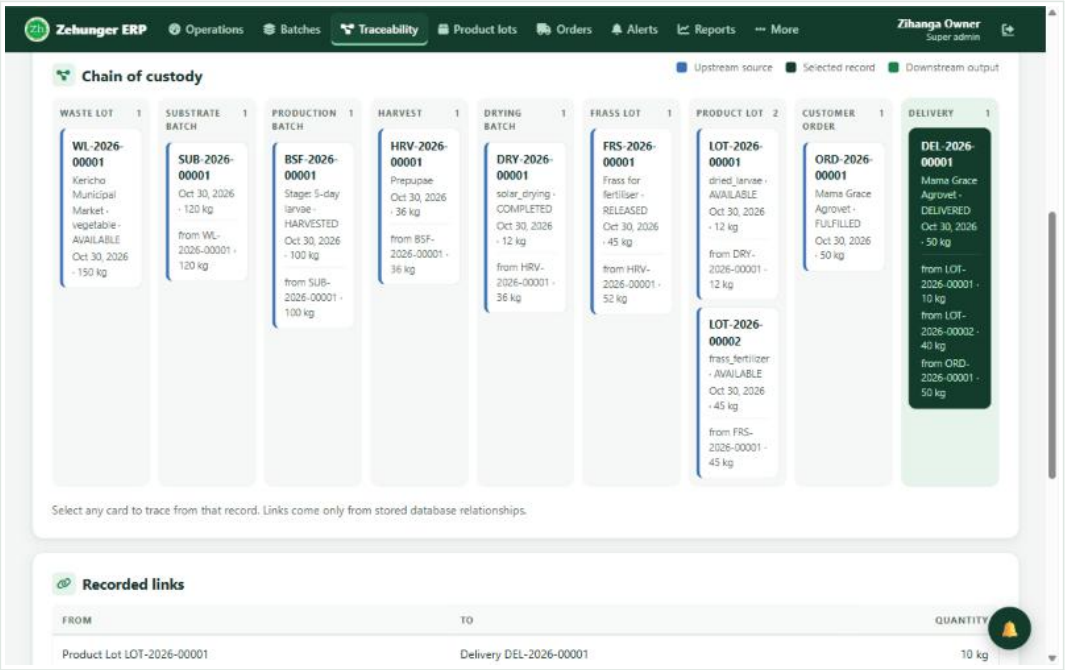
STEP 10

10 The owner's view

Who: Super admin (owner) · **Where:** Traceability, Operations, Batch 360, Reports

10.1 Trace any record

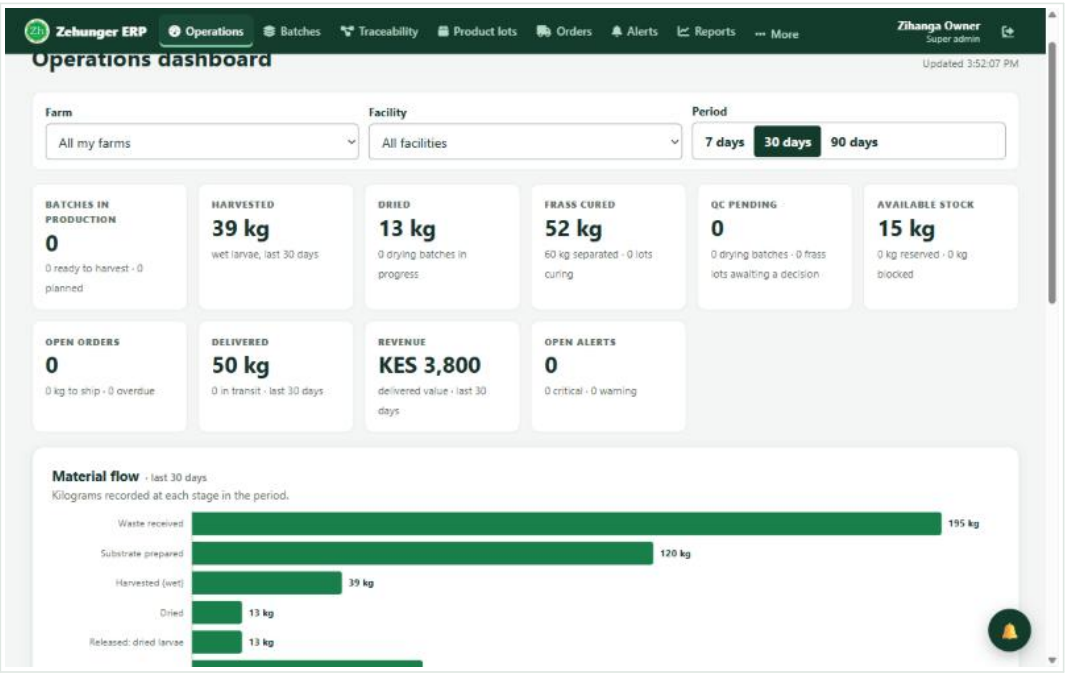
Traceability → enter **DEL-2026-00001** → **Trace**.



Chain of custody from waste lot to delivery

The delivery traces back through **9 upstream records**: both product lots, the drying batch and the frass lot, the harvest, the production batch, the substrate and the waste lot from *Kericho Municipal Market*. Every link is a stored record, never a guess. Click any card to trace from that record.

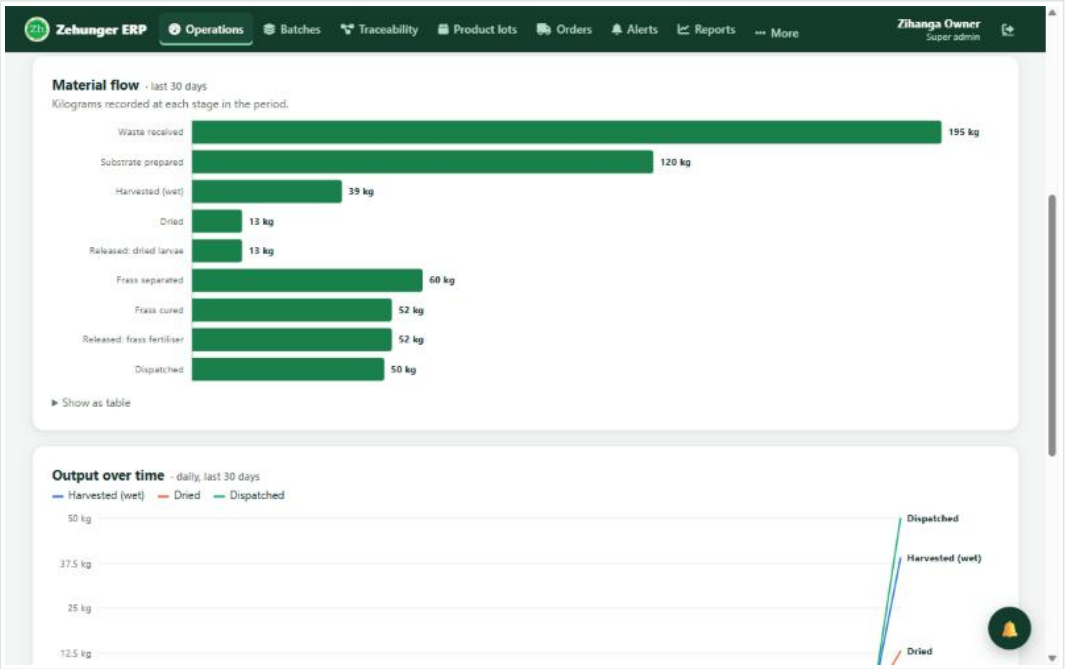
10.2 Operations dashboard



Operations dashboard tiles

Tile	Walkthrough value
Harvested	39 kg wet larvae
Dried	13 kg
Frass cured	52 kg (60 kg separated)

Tile	Walkthrough value
QC pending	0
Available stock	15 kg (3 kg larvae + 12 kg frass)
Delivered	50 kg
Revenue	KES 3,800
Open alerts	0

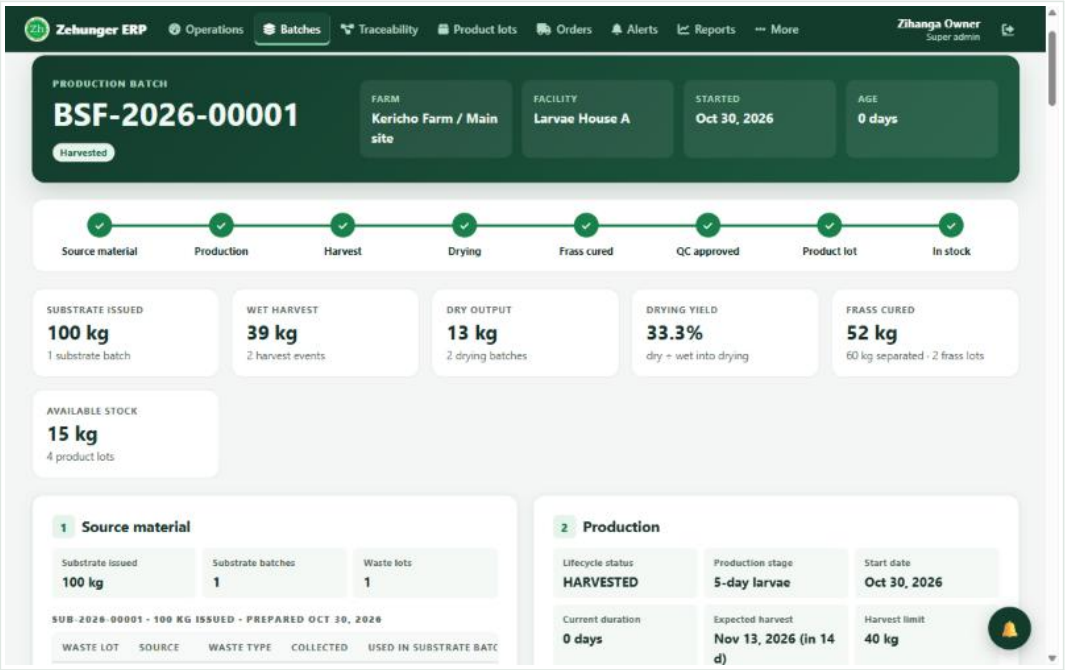


Material flow: both products, stage by stage

Material flow shows each stage in kilograms: waste 195 → substrate 120 → harvested 39 → dried 13 / released 13, and frass separated 60 → cured 52 → released 52 → dispatched 50. Filter by farm, facility and 7, 30 or 90 days.

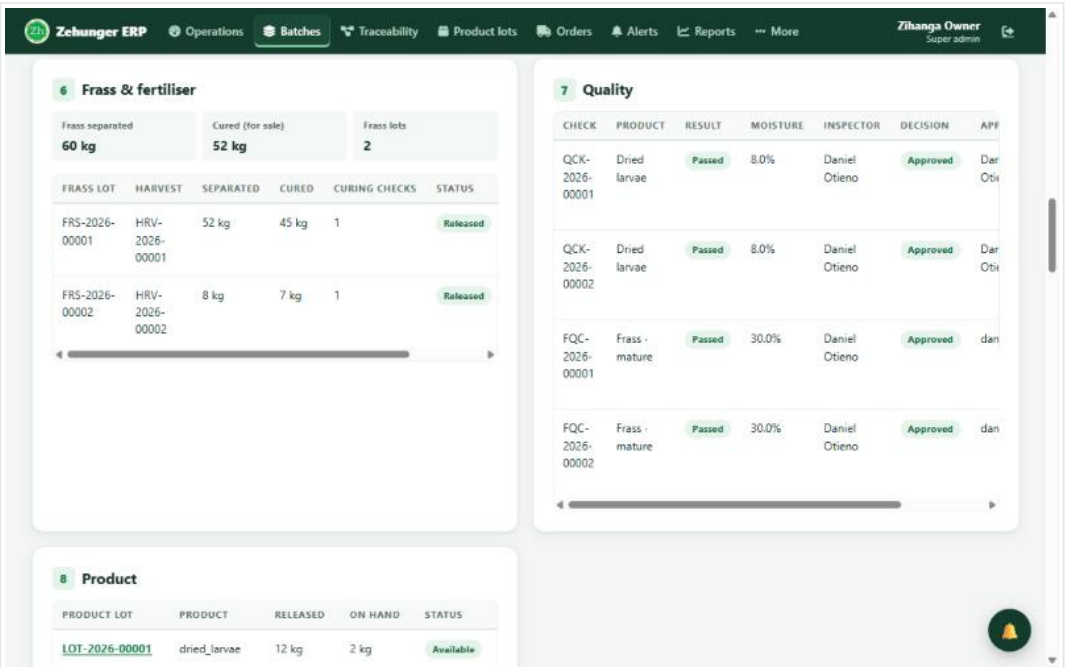
10.3 Batch 360

Batches → *Batch 360* → **BSF-2026-00001**: one batch, end to end, on one page.



Batch 360 header, lifecycle and key figures

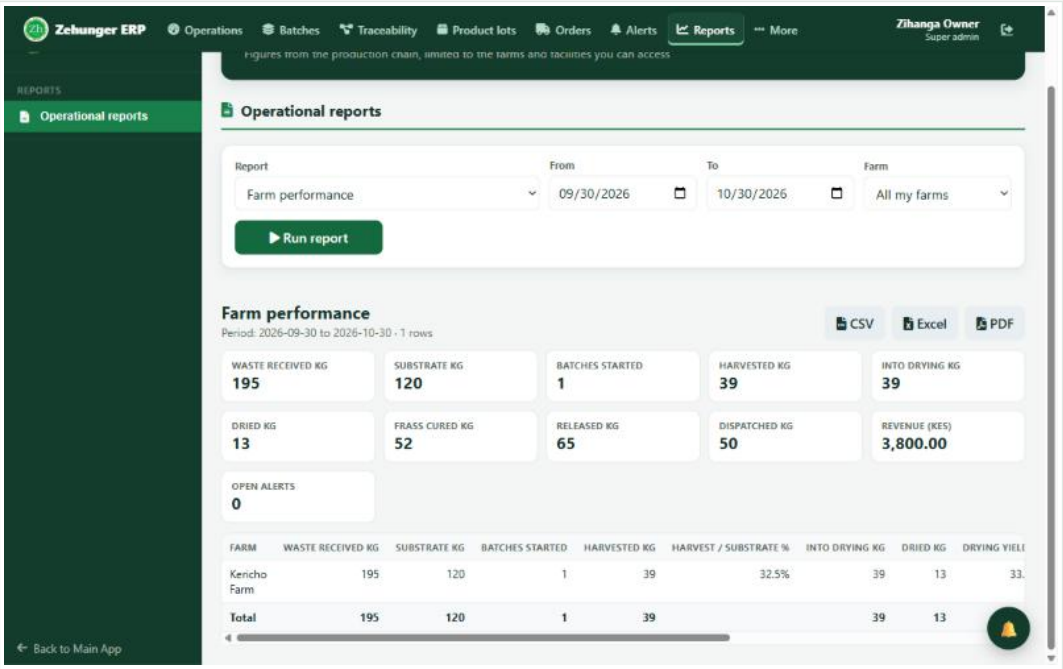
- The **lifecycle strip** shows each stage done: Source material, Production, Harvest, Drying, Frass cured, QC approved, Product lot, In stock.
- **Key figures:** 100 kg substrate, 39 kg harvest, 13 kg dry (33.3% yield), 52 kg frass cured, 15 kg in stock.
- Numbered cards cover source material, production, operations, harvest, processing, **frass & fertiliser**, quality, product, inventory, traceability (including customers) and a full audit trail.



Batch 360: frass and quality cards

10.4 Reports

Reports → choose a report, period and farm → **Run report**. Every report exports to **CSV, Excel or PDF**.



Farm performance report

Group	Reports
Production	Waste received, Substrate preparation, Production batches, Feeding activity, Harvest, Farm operation logs
Processing	Drying, **Frass & fertiliser**, Quality control
Stock & sales	Current stock, Customer orders, Deliveries
Traceability	Product lineage
Overview	Farm performance

Farm performance for Kericho Farm: waste 195 kg, substrate 120 kg, harvested 39 kg (32.5% of substrate), dried 13 kg (33.3% drying yield), frass cured 52 kg, released 65 kg, dispatched 50 kg, revenue KES 3,800.

Rules the system enforces

A summary for questions from the owners:

1. **Traceability is built in.** Every record links to its source; links are stored, never inferred.
2. **Quality gate.** No product reaches stock without a passed check and an approval; the database refuses it.
3. **Mass balance.** You cannot use more waste, substrate or harvest than exists, or harvest above a batch's limit.
4. **No future dates** on records.
5. **Location access.** People see only their farms, sites and facilities; sales and money need farm-level access.
6. **Roles.** Each action needs the right role: operators record work, QC officers decide quality, sales officers sell.
7. **Full audit trail.** Every change records who did it and when.
8. **Nothing is deleted once used.** Places and records are deactivated or reversed, so history stays complete.

Suggested presentation script (about 25 minutes)

Time	Step	Say this
0–2 min	Introduction	"One batch, from market waste to a customer's shop, with every kilogram accounted for."
2–4	Step 0 and sign-in	Farm set-up and roles: each person sees only their own work.
4–7	Steps 1–2	Waste in, quality gate (quarantine), substrate links back to waste.
7–10	Steps 3–5	Production batch, daily records, harvest with frass as a second product.
10–13	Steps 6–7	Drying at 3:1; frass cured into fertiliser.
13–16	Step 8	QC: check, decide, release — the system blocks anything unapproved.
16–20	Step 9	Customer, order with prices, reserve, dispatch, delivery and revenue.
20–24	Step 10	Trace a delivery back to the market; dashboard; Batch 360; reports.
24–25	Close	What comes next (see below), questions.

Questions the audience may ask

Can a farmer fake a harvest figure? Every harvest is capped by the batch's harvest limit, linked to its batch and signed with the user and time. The audit trail shows any change.

What if a product fails QC? It is rejected with a reason and never reaches stock. A failed drying output can be re-checked while the batch waits for QC.

Can we sell frass? Yes. Frass is separated at harvest, cured, quality-checked and released as *frass fertiliser*, and it is ordered, delivered and traced exactly like dried larvae.

How do we see profit? Today the system records revenue (delivered kg × price). A full double-entry sales ledger with invoices, payments and customer balances is planned for the next phase.

Does it work on a phone? Yes. The Farmer Portal and all forms are designed for phones; operators record work in the field.

Where is the data kept? In a managed PostgreSQL database (Neon) that can be restored to an earlier point in time; the application runs on a hosted server (Render). Every working page requires sign-in.

Reference numbers

Prefix	Record
WL-	Waste lot
SUB-	Substrate batch
BSF-	Production batch
EVT-	Production event (feeding, monitoring, health, harvest)
HRV-	Harvest
FRS-	Frass lot
DRY-	Drying batch
QCK-	Quality check (dried larvae)
QCA-	Quality approval (dried larvae)
FQC-	Frass quality check
LOT-	Product lot (stock)
ORD-	Customer order
DEL-	Delivery

What comes next

- **Sales ledger:** invoices, payments (M-Pesa, bank, cash), customer balances and a double-entry ledger.
- **Waste collection page:** a redesigned waste intake screen (already built, held for after the presentation).
- **Cost tracking:** cost per kilogram, to report margins.
- **Hatchery and breeding links:** connecting egg and fly records to production batches.

Zehunger Solutions — Circular Economy ERP. Phase One walkthrough guide, 30 October 2026.